BlockChain Experiment 09

Aim: To implement a Private Ethereum Blockchain using Geth.

Theory:

Ethereum is an open-source, decentralized blockchain platform that enables developers to build and deploy smart contracts and decentralized applications (DApps). Unlike public blockchains such as the Ethereum mainnet, a Private Ethereum Blockchain is restricted to specific participants, making it ideal for development, testing, and enterprise applications where confidentiality, control, and performance are essential.

Geth (Go-Ethereum) is one of the most widely used Ethereum clients, written in the Go programming language. It allows users to run an Ethereum node, mine Ether, and interact with the blockchain through JSON-RPC, command-line, or JavaScript consoles.

Setting up a Private Ethereum Network involves creating a customized blockchain environment isolated from the public Ethereum network. It enables developers to test smart contracts, consensus mechanisms, and network behavior without incurring gas costs or depending on public validators.

Steps Involved:

1. **Choosing a Network ID:**
2. Each private blockchain must have a unique network ID to differentiate it from public or other private networks.
3. **Choosing a Consensus Algorithm:**
4. Ethereum supports different consensus mechanisms such as **Proof of Work (PoW)** and **Proof of Authority (PoA)**. In private networks, **Clique (PoA)** is often used for faster block confirmation.
5. **Creating a Genesis Block:**
6. The genesis block is the first block of the blockchain, defining network parameters such as difficulty, gas limit, initial account balances, and consensus settings.
7. **Initializing the Geth Database:**
8. The geth init command initializes the blockchain using the **genesis.json** file, preparing the data directory for node operation.
9. **Setting up Networking:**
10. Nodes communicate using **enodes** (Ethereum Node URLs) that contain IP addresses and public keys for peer discovery and connection.
11. **Running the Member Nodes:**
12. Each participant runs a Geth node that connects to the private network and participates in block validation and transaction processing.
13. **Running a Signer (in Clique):**
14. In the **Clique consensus**, designated **signers** validate and sign blocks. These accounts are pre-configured in the genesis file to ensure trusted block production.

Note: Download the genesis file and edit the account details, including the public keys of all participating peers in the network.

Implementation:

```
quantum_pulse@LAPTOP-DVBOTNG9:~$ cd private_ethereum_setup
quantum_pulse@LAPTOP-DVBOTNG9:~/private_ethereum_setup$ geth init --datadir node1 genesis.json
INFO [10-10|12:46:11.508] Maximum peer count                       ETH=50 LES=0 total=50
INFO [10-10|12:46:11.513] Smartcard socket not found, disabling    err="stat /run/pcscd/pcscd.comm: no such fil
e or directory"
INFO [10-10|12:46:11.531] Set global gas cap                       cap=50,000,000
INFO [10-10|12:46:11.532] Initializing the KZG library             backend=gokzg
INFO [10-10|12:46:11.622] Using pebble as the backing database
INFO [10-10|12:46:11.623] Allocated cache and file handles         database=/home/quantum_pulse/private_ethereu
m_setup/node1/geth/chaindata cache=16.00MiB handles=16
INFO [10-10|12:46:11.686] Opened ancient database                  database=/home/quantum_pulse/private_ethereu
m_setup/node1/geth/chaindata/ancient/chain readonly=false
INFO [10-10|12:46:11.704] Successfully wrote genesis state         database=chaindata hash=a66e60..463653
INFO [10-10|12:46:11.704] Using pebble as the backing database
INFO [10-10|12:46:11.705] Allocated cache and file handles         database=/home/quantum_pulse/private_ethereu
m_setup/node1/geth/lightchaindata cache=16.00MiB handles=16
INFO [10-10|12:46:11.753] Opened ancient database                  database=/home/quantum_pulse/private_ethereu
m_setup/node1/geth/lightchaindata/ancient/chain readonly=false
INFO [10-10|12:46:11.753] Freezer shutting down
INFO [10-10|12:46:11.766] Successfully wrote genesis state         database=lightchaindata hash=a66e60..463653
```

```
quantum_pulse@LAPTOP-DVBOTNG9:~/private_ethereum_setup$ geth init --datadir node2 genesis.json
INFO [10-10|12:46:16.738] Maximum peer count                       ETH=50 LES=0 total=50
INFO [10-10|12:46:16.740] Smartcard socket not found, disabling    err="stat /run/pcscd/pcscd.comm: no such fil
e or directory"
INFO [10-10|12:46:16.754] Set global gas cap                       cap=50,000,000
INFO [10-10|12:46:16.755] Initializing the KZG library             backend=gokzg
INFO [10-10|12:46:16.839] Using pebble as the backing database
INFO [10-10|12:46:16.839] Allocated cache and file handles         database=/home/quantum_pulse/private_ethereu
m_setup/node2/geth/chaindata cache=16.00MiB handles=16
INFO [10-10|12:46:16.914] Opened ancient database                  database=/home/quantum_pulse/private_ethereu
m_setup/node2/geth/chaindata/ancient/chain readonly=false
INFO [10-10|12:46:16.935] Successfully wrote genesis state         database=chaindata hash=a66e60..463653
INFO [10-10|12:46:16.937] Using pebble as the backing database
INFO [10-10|12:46:16.937] Allocated cache and file handles         database=/home/quantum_pulse/private_ethereu
m_setup/node2/geth/lightchaindata cache=16.00MiB handles=16
INFO [10-10|12:46:16.998] Opened ancient database                  database=/home/quantum_pulse/private_ethereu
m_setup/node2/geth/lightchaindata/ancient/chain readonly=false
INFO [10-10|12:46:17.013] Successfully wrote genesis state         database=lightchaindata hash=a66e60..463653
quantum_pulse@LAPTOP-DVBOTNG9:~/private_ethereum_setup$ bootnode -genkey boot.key
quantum_pulse@LAPTOP-DVBOTNG9:~/private_ethereum_setup$ bootnode -nodekey boot.key -verbosity 9 -addr :30305
enode://15a054292fefbfb795656b041553a7b57544a3cae7d786c3d6edbd40edf860ff243b9970c210b6573495d0709d5d62fccd97ba8
41e0629d8252789be919c983e@127.0.0.1:0?discport=30305
Note: you're using cmd/bootnode, a developer tool.
We recommend using a regular node as bootstrap node for production deployments.
INFO [10-10|12:47:10.297] New local node record                    seq=1,760,080,630,295 id=61c8b037b2deca33 ip
="invalid IP" udp=0 tcp=0
TRACE[10-10|12:48:17.934] << PING/v4                               id=12340f17ea8deb0d addr=127.0.0.1:30306 err
```

```
quantum_pulse@LAPTOP-DVBOTNG9:~/private_ethereum_setup$ bootnode -genkey boot.key
quantum_pulse@LAPTOP-DVBOTNG9:~/private_ethereum_setup$ bootnode -nodekey boot.key -verbosity 9 -addr :30305
enode://15a054292fefbfb795656b041553a7b57544a3cae7d786c3d6edbd40edf860ff243b9970c210b6573495d0709d5d62fccd97ba8
41e0629d8252789be919c983e@127.0.0.1:0?discport=30305
Note: you're using cmd/bootnode, a developer tool.
We recommend using a regular node as bootstrap node for production deployments.
INFO [10-10|12:47:10.297] New local node record                    seq=1,760,080,630,295 id=61c8b037b2deca33 ip
="invalid IP" udp=0 tcp=0
TRACE[10-10|12:48:17.934] << PING/v4                               id=12340f17ea8deb0d addr=127.0.0.1:30306 err
=<nil>
TRACE[10-10|12:48:17.936] >> PONG/v4                               id=12340f17ea8deb0d addr=127.0.0.1:30306 err
=<nil>
TRACE[10-10|12:48:17.939] >> PING/v4                               id=12340f17ea8deb0d addr=127.0.0.1:30306 err
=<nil>
TRACE[10-10|12:48:17.940] << PING/v4                               id=12340f17ea8deb0d addr=127.0.0.1:30306 err
=<nil>
TRACE[10-10|12:48:17.940] >> PONG/v4                               id=12340f17ea8deb0d addr=127.0.0.1:30306 err
=<nil>
TRACE[10-10|12:48:17.941] >> PING/v4                               id=12340f17ea8deb0d addr=127.0.0.1:30306 err
=<nil>
TRACE[10-10|12:48:17.941] << PING/v4                               id=12340f17ea8deb0d addr=127.0.0.1:30306 err
=<nil>
TRACE[10-10|12:48:17.942] >> PONG/v4                               id=12340f17ea8deb0d addr=127.0.0.1:30306 err
=<nil>
TRACE[10-10|12:48:17.943] >> PING/v4                               id=12340f17ea8deb0d addr=127.0.0.1:30306 err
=<nil>
```

```
quantum_pulse@LAPTOP-DVBOTNG9:~/private_ethereum_setup$ geth --datadir node1 --port 30306 --bootnodes enode://1
5a054292fefbfb795656b041553a7b57544a3cae7d786c3d6edbd40edf860ff243b9970c210b6573495d0709d5d62fccd97ba841e0629d8
252789be919c983e@127.0.0.1:0?discport=30305 --networkid 123454321 --unlock 0xA897DC86D6331fEA9a0C3Ca5b43240ac5A
dc0418 --password node1/password.txt --authrpc.port 8551 --miner.etherbase 0xA897DC86D6331fEA9a0C3Ca5b43240ac5A
dc0418 --mine
INFO [10-10|12:48:17.204] Maximum peer count                       ETH=50 LES=0 total=50
INFO [10-10|12:48:17.207] Smartcard socket not found, disabling    err="stat /run/pcscd/pcscd.comm: no such fil
e or directory"
INFO [10-10|12:48:17.217] Set global gas cap                       cap=50,000,000
INFO [10-10|12:48:17.219] Initializing the KZG library             backend=gokzg
INFO [10-10|12:48:17.346] Allocated trie memory caches             clean=154.00MiB dirty=256.00MiB
INFO [10-10|12:48:17.347] Using pebble as the backing database
INFO [10-10|12:48:17.347] Allocated cache and file handles         database=/home/quantum_pulse/private_ethereu
m_setup/node1/geth/chaindata cache=512.00MiB handles=524,288
INFO [10-10|12:48:17.402] Opened ancient database                  database=/home/quantum_pulse/private_ethereu
m_setup/node1/geth/chaindata/ancient/chain readonly=false
INFO [10-10|12:48:17.407] Initialising Ethereum protocol           network=123,454,321 dbversion=8
INFO [10-10|12:48:17.413]
INFO [10-10|12:48:17.413] ----------------------------------------------------------------------------------
-------------------------------------------------------------------
```

```
quantum_pulse@LAPTOP-DVBOTNG9:~/private_ethereum_setup$ geth --datadir node2 --port 30307 --bootnodes en
ode://15a054292fefbfb795656b041553a7b57544a3cae7d786c3d6edbd40edf860ff243b9970c210b6573495d0709d5d62fccd
97ba841e0629d8252789be919c983e@127.0.0.1:0?discport=30305 --networkid 123454321 --unlock 0xDeD436D1dCb78
9E546682e01fA5e2827Cce0E32C --password node2/password.txt --authrpc.port 8552
INFO [10-10|12:48:44.267] Maximum peer count                       ETH=50 LES=0 total=50
INFO [10-10|12:48:44.269] Smartcard socket not found, disabling    err="stat /run/pcscd/pcscd.comm: no s
uch file or directory"
INFO [10-10|12:48:44.278] Set global gas cap                       cap=50,000,000
INFO [10-10|12:48:44.278] Initializing the KZG library             backend=gokzg
INFO [10-10|12:48:44.364] Allocated trie memory caches             clean=154.00MiB dirty=256.00MiB
INFO [10-10|12:48:44.364] Using pebble as the backing database
INFO [10-10|12:48:44.364] Allocated cache and file handles         database=/home/quantum_pulse/private_
ethereum_setup/node2/geth/chaindata cache=512.00MiB handles=524,288
INFO [10-10|12:48:44.413] Opened ancient database                  database=/home/quantum_pulse/private_
ethereum_setup/node2/geth/chaindata/ancient/chain readonly=false
INFO [10-10|12:48:44.423] Initialising Ethereum protocol           network=123,454,321 dbversion=8
INFO [10-10|12:48:44.427]
INFO [10-10|12:48:44.427] ----------------------------------------------------------------------------
-----------------------------------------------------------------------
INFO [10-10|12:48:44.427] Chain ID:  12345 (unknown)
```

```
quantum_pulse@LAPTOP-DVBOTNG9:~/private_ethereum_setup$ geth --datadir node2 --port 30307 --bootnodes en
ode://15a054292fefbfb795656b041553a7b57544a3cae7d786c3d6edbd40edf860ff243b9970c210b6573495d0709d5d62fccd
97ba841e0629d8252789be919c983e@127.0.0.1:0?discport=30305 --networkid 123454321 --unlock 0xDeD436D1dCb78
9E546682e01fA5e2827Cce0E32C --password node2/password.txt --authrpc.port 8552
INFO [10-10|12:48:44.267] Maximum peer count                       ETH=50 LES=0 total=50
INFO [10-10|12:48:44.269] Smartcard socket not found, disabling    err="stat /run/pcscd/pcscd.comm: no s
uch file or directory"
INFO [10-10|12:48:44.278] Set global gas cap                       cap=50,000,000
INFO [10-10|12:48:44.278] Initializing the KZG library             backend=gokzg
INFO [10-10|12:48:44.364] Allocated trie memory caches             clean=154.00MiB dirty=256.00MiB
INFO [10-10|12:48:44.364] Using pebble as the backing database
INFO [10-10|12:48:44.364] Allocated cache and file handles         database=/home/quantum_pulse/private_
ethereum_setup/node2/geth/chaindata cache=512.00MiB handles=524,288
INFO [10-10|12:48:44.413] Opened ancient database                  database=/home/quantum_pulse/private_
ethereum_setup/node2/geth/chaindata/ancient/chain readonly=false
INFO [10-10|12:48:44.423] Initialising Ethereum protocol           network=123,454,321 dbversion=8
INFO [10-10|12:48:44.427]
INFO [10-10|12:48:44.427] ----------------------------------------------------------------------------
-----------------------------------------------------------------------
INFO [10-10|12:48:44.427] Chain ID:  12345 (unknown)
```

```
quantum_pulse@LAPTOP-DVBOTNG9:~/private_ethereum_setup$ geth attach node1/geth.ipc
Welcome to the Geth JavaScript console!

instance: Geth/v1.12.0-stable-e501b3b0/linux-amd64/go1.20.3
coinbase: 0xa897dc86d6331fea9a0c3ca5b43240ac5adc0418
at block: 647 (Fri Oct 10 2025 12:49:09 GMT+0530 (IST))
 datadir: /home/quantum_pulse/private_ethereum_setup/node1
 modules: admin:1.0 clique:1.0 debug:1.0 engine:1.0 eth:1.0 miner:1.0 net:1.0 rpc:1.0 txpool:1.0 web3:1.0

To exit, press ctrl-d or type exit
> net.peerCount
1
> net.peerCount
1
> eth.accounts[0]
"0xa897dc86d6331fea9a0c3ca5b43240ac5adc0418"
```

Conclusion

We successfully created and configured a private Ethereum blockchain using Geth in this experiment, gaining valuable hands-on experience with the core components that power decentralized systems. By setting up a customized genesis block, we defined the initial state and rules of the blockchain, which is a crucial step in establishing a controlled network environment. Initializing and connecting multiple nodes further helped in understanding how peer-to-peer communication works, allowing nodes to synchronize data and maintain a consistent distributed ledger.

The implementation of the Clique consensus algorithm provided insight into how consensus mechanisms ensure agreement among network participants. Unlike energy-intensive mining processes, Clique offers a more efficient proof-of-authority approach, making it ideal for private networks where participants are known and trusted. This not only improves performance but also highlights how different consensus models can be chosen based on specific use cases and requirements.Moreover, this setup enabled us to simulate a secure and controlled blockchain environment suitable for testing, experimentation, and development without the unpredictability of public networks. It demonstrated how developers can deploy smart contracts, monitor transactions, and analyze network behavior in a risk-free setting. Overall, this experiment strengthened our understanding of fundamental blockchain concepts such as network configuration, consensus, decentralization, and node management, while also providing practical exposure to real-world blockchain implementation.